

Course 5251

5 Credits: No Credit

Program Core

Source-grounded

Visual Programming

□ To provide an understanding of the programming concepts using VB.NET

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5251 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5251.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Computer Application and Business Management
Course code	5251
Course title	Visual Programming
Semester	5 Credits: No Credit
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	No Credit
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

38 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide an understanding of the programming concepts using VB.NET Programming Language.
- To understand the .NET framework for creating web applications.

Course outcomes

CO	Official outcome	Study level
CO1	13 Understanding constants, operators & expressions.	See official syllabus
CO2	Summarize control structures and arrays 10 Understanding Explain the concept of windows programming,	See official syllabus
CO3	20 Understanding events and exceptions	See official syllabus
CO4	Outline ADO.NET and ASP.NET 15 Understanding Series Exam 2	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	expressions.	8	As specified
Module 2	Summarize control structures and Arrays.	8	As specified
Module 3	Explain the concept of windows programming, events and exceptions.	14	As specified
Module 4	Outline ADO.NET and ASP.NET.	8	As specified

MODULE 1

expressions.

This module is presented from the official syllabus scope for Visual Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: expressions.
- Official duration: As specified
- Official topic entries captured: 8

- The full source excerpt is preserved below for coverage checking.

Define Visual Programming 1 Understanding List advantages and disadvantages of visual

Define Visual Programming 1 Understanding List advantages and disadvantages of visual is an official topic in Module 1 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define Visual Programming 1 Understanding List advantages and disadvantages of visual using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define visual programming 1 understanding list advantages and disadvantages of visual should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define Visual Programming 1 Understanding List advantages and disadvantages of visual means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Define Visual Programming 1 Understanding List advantages and disadvantages of visual programming. definition/meaning. Advantages and disadvantages, steps, application of visual programming. Technical terms English-Malayalam.

1 Understanding programming Familiarize .NET Framework and explain .NET

1 Understanding programming Familiarize .NET Framework and explain .NET is an official topic in Module 1 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding programming Familiarize .NET Framework and explain .NET using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding programming familiarize .net framework and explain .net should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding programming Familiarize .NET Framework and explain .NET means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1- 1 Understanding programming Familiarize .NET Framework and explain .NET definition/meaning
definition/meaning. definition/meaning definition/meaning, steps, application definition/meaning
definition/meaning. Technical terms English- definition/meaning definition/meaning.

2 Understanding Framework architecture

2 Understanding Framework architecture is an official topic in Module 1 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Framework architecture using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding framework architecture should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Framework architecture means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-2 Understanding Framework architecture
ഘടനാപരമായ ഘടനാ definition/meaning ഘടനാപരമായ ഘടനാ. ഘടനാ ഘടനാ ഘടനാ,
steps, application ഘടനാ ഘടനാപരമായ ഘടനാ. Technical terms English-ഘ ഘടനാ
ഘടനാപരമായ ഘടനാ.

Understand CLR and .Net Class Library

Understand CLR and .Net Class Library is an official topic in Module 1 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand CLR and .Net Class Library using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand clr and .net class library should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand CLR and .Net Class Library means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Understand CLR and .Net Class Library
പ്രവർത്തനം നിർവ്വചന/അർത്ഥം നിർവ്വചിക്കുക. പ്രവർത്തനം നിർവ്വചിക്കുക,
steps, application നിർവ്വചനം നിർവ്വചിക്കുക. Technical terms English-ൽ നിർവ്വചിക്കുക.
നിർവ്വചിക്കുക.

Compare .NET Languages VB, C# 1 Understanding Explain variables and constants with examples,

Compare .NET Languages VB, C# 1 Understanding Explain variables and constants with examples, is an official topic in Module 1 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compare .NET Languages VB, C# 1 Understanding Explain variables and constants with examples, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compare .net languages vb, c# 1 understanding explain variables and constants with examples, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compare .NET Languages VB, C# 1 Understanding Explain variables and constants with examples, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Compare .NET Languages VB, C# 1

Understanding Explain variables and constants with examples, definition/meaning definition/meaning steps, application Technical terms English-

2 Understanding understand data types and declaration

2 Understanding understand data types and declaration is an official topic in Module 1 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding understand data types and declaration using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding understand data types and declaration should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding understand data types and declaration means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Understanding understand data types and declaration ഓരോന്നിന്റെയും definition/meaning ഓരോന്നിന്റെയും. ഓരോന്നിന്റെയും steps, application ഓരോന്നിന്റെയും. Technical terms English-ൽ ഓരോന്നിന്റെയും.

Explain arithmetic, relational and logical operators 2 Understanding Understand operator precedence, expressions and

Explain arithmetic, relational and logical operators 2 Understanding Understand operator precedence, expressions and is an official topic in Module 1 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain arithmetic, relational and logical operators 2 Understanding Understand operator precedence, expressions and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain arithmetic, relational and logical operators 2 understanding understand operator precedence, expressions and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain arithmetic, relational and logical operators 2 Understanding Understand operator precedence, expressions and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-
 Explain arithmetic, relational and logical operators
 2 Understanding Understand operator precedence, expressions and
 definition/meaning .
 steps, application . Technical terms English-
 .

2 Understanding managed code Visual Programming - .NET Framework - CLR - .NET Class Library - .NET Languages - Variables - Constants - Data Types - Operators - Expressions.

2 Understanding managed code Visual Programming - .NET Framework - CLR - .NET Class Library - .NET Languages - Variables - Constants - Data Types - Operators - Expressions. is an official topic in Module 1 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding managed code Visual Programming - .NET Framework - CLR - .NET Class Library - .NET Languages - Variables - Constants - Data Types - Operators - Expressions. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding managed code visual programming - .net framework - clr - .net class library - .net languages - variables - constants - data types - operators - expressions. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding managed code Visual Programming - .NET Framework - CLR - .NET Class Library - .NET Languages - Variables - Constants - Data Types - Operators - Expressions. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-2 Understanding managed code Visual Programming - .NET Framework - CLR - .NET Class Library - .NET Languages - Variables - Constants - Data Types - Operators - Expressions. പദപരിചയം പദപരിചയം definition/meaning പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-ലേക്ക് പദപരിചയം പദപരിചയം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

expressions.		
M1.01	Define Visual Programming	1
Understanding	List advantages and disadvantages of visual	
M1.02		1
Understanding	programming	
	Familiarize .NET Framework and explain .NET	
M1.03		2
Understanding	Framework architecture	
M1.04	Understand CLR and .Net Class Library	2
Understanding		
M1.05	Compare .NET Languages VB, C#	1
Understanding	Explain variables and constants with examples,	
M1.06		2
Understanding	understand data types and declaration	
M1.07	Explain arithmetic, relational and logical operators	2
Understanding	Understand operator precedence, expressions and	
M1.08		2
Understanding	managed code	

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Summarize control structures and Arrays.

This module is presented from the official syllabus scope for Visual Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize control structures and Arrays.
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

Understand If - Then, If - Then - Else Statements

Understand If - Then, If - Then - Else Statements is an official topic in Module 2 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand If - Then, If - Then - Else Statements using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand if - then, if - then - else statements should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand If - Then, If - Then - Else Statements means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Understand If - Then, If - Then - Else Statements
 definition/meaning . steps, application . Technical terms English-
 .

Understand Select Case Statement

Understand Select Case Statement is an official topic in Module 2 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand Select Case Statement using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand select case statement should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand Select Case Statement means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-☐☐ Understand Select Case Statement

ৱিকিপিডিয়াৰ পৰা definition/meaning ৰচনা কৰা হৈছে. ৱিকিপিডিয়াৰ পৰা steps, application ৰচনা কৰা হৈছে. Technical terms English-ৰ ৱিকিপিডিয়াৰ পৰা ৰচনা কৰা হৈছে.

Understand Do While - Loop Statement

Understand Do While - Loop Statement is an official topic in Module 2 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand Do While - Loop Statement using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand do while - loop statement should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand Do While - Loop Statement means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Do While - Loop Statement

Do While Loop statement definition/meaning. Do While Loop statement steps, application. Technical terms English- Malayalam.

Understand For - Next Statement

Understand For - Next Statement is an official topic in Module 2 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand For - Next Statement using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand for - next statement should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand For - Next Statement means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Understand For - Next Statement
 definition/meaning . . . , steps, application . Technical terms English-
 .

Understand exit, continue and break statements

Understand exit, continue and break statements is an official topic in Module 2 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand exit, continue and break statements using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand exit, continue and break statements should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand exit, continue and break statements means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Understand exit, continue and break statements
 definition/meaning .
 steps, application
 Technical terms
 English-
 .

Understand Single Dimensional Arrays

Understand Single Dimensional Arrays is an official topic in Module 2 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand Single Dimensional Arrays using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand single dimensional arrays should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand Single Dimensional Arrays means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓരോ Understand Single Dimensional Arrays

ഓരോരേഖാമൂലം ഓരോരേഖാ definition/meaning ഓരോരേഖാമൂലം. ഓരോരേഖാ ഓരോരേഖാ ഓരോരേഖാ, steps, application ഓരോരേഖാ ഓരോരേഖാമൂലം ഓരോരേഖാ. Technical terms English-ഓരോരേഖാ ഓരോരേഖാമൂലം.

Understand Multi-Dimensional Arrays

Understand Multi-Dimensional Arrays is an official topic in Module 2 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand Multi-Dimensional Arrays using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand multi-dimensional arrays should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand Multi-Dimensional Arrays means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Understand Multi-Dimensional Arrays

മൾട്ടി-ഡൈമൻഷണൽ അറേയ്ക്ക് definition/meaning എന്താണ്. അറേയ്ക്ക് ഏതെങ്കിലും steps, application എന്തെങ്കിലും ഉദാഹരണങ്ങൾ. Technical terms English-ൽ എഴുതുക.

Understand Object Arrays 1 Understanding Control Statements: If Then - If Then Else - Select Case - Looping Statements: Do While - For - Exit - Continue - Arrays - Single Dimensional - Multidimensional - Object Arrays.

Understand Object Arrays 1 Understanding Control Statements: If Then - If Then Else - Select Case - Looping Statements: Do While - For - Exit - Continue - Arrays - Single Dimensional - Multidimensional - Object Arrays. is an official topic in Module 2 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand Object Arrays 1 Understanding Control Statements: If Then - If Then Else - Select Case - Looping Statements: Do While - For - Exit - Continue - Arrays - Single Dimensional - Multidimensional - Object Arrays. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand object arrays 1 understanding control statements: if then - if then else - select case - looping statements: do while - for - exit - continue - arrays - single dimensional - multidimensional - object arrays. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand Object Arrays 1 Understanding Control Statements: If Then - If Then Else - Select Case - Looping Statements: Do While - For - Exit - Continue - Arrays - Single Dimensional - Multidimensional - Object Arrays. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓബ്ജക്ട് അറേയ്സ് 1 Understanding Control Statements: If Then - If Then Else - Select Case - Looping Statements: Do While - For - Exit - Continue - Arrays - Single Dimensional - Multidimensional - Object Arrays. ഓബ്ജക്ട് അറേയ്സ് definition/meaning ഓബ്ജക്ട് അറേയ്സ്. ഓബ്ജക്ട് അറേയ്സ് ഓബ്ജക്ട് അറേയ്സ്, steps, application ഓബ്ജക്ട് അറേയ്സ് ഓബ്ജക്ട് അറേയ്സ്. Technical terms English-ഓബ്ജക്ട് അറേയ്സ് ഓബ്ജക്ട് അറേയ്സ്.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Summarize control structures and Arrays.

M2.01	Understand If - Then, If - Then - Else Statements	2
Understanding		
M2.02	Understand Select Case Statement	1
Understanding		
M2.03	Understand Do While - Loop Statement	1
Understanding		
M2.04	Understand For - Next Statement	1
Understanding		
M2.05	Understand exit, continue and break statements	1
Understanding		
M2.06	Understand Single Dimensional Arrays	2
Understanding		
M2.07	Understand Multi-Dimensional Arrays	1
Understanding		
M2.08	Understand Object Arrays	1
Understanding		
	Series Test-I	1

Contents:

Control Statements: If Then - If Then Else - Select Case - Looping

Statements: Do While - For - Exit - Continue - Arrays - Single Dimensional -

Multidimensional - Object Arrays.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Explain the concept of windows programming, events and exceptions.

This module is presented from the official syllabus scope for Visual Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the concept of windows programming, events and exceptions.
- Official duration: As specified
- Official topic entries captured: 14
- The full source excerpt is preserved below for coverage checking.

Describe the procedure of creating Window Forms

Describe the procedure of creating Window Forms is an official topic in Module 3 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the procedure of creating Window Forms using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the procedure of creating window forms should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the procedure of creating Window Forms means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Describe the procedure of creating Window Forms
 definition/meaning .
 , steps, application . Technical terms English-
 .

Explain Form Attributes 1 Understanding Define TextBox, ComboBox, ListBox, CheckBox

Explain Form Attributes 1 Understanding Define TextBox, ComboBox, ListBox, CheckBox is an official topic in Module 3 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Form Attributes 1 Understanding Define TextBox, ComboBox, ListBox, CheckBox using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain form attributes 1 understanding define textbox, combobox, listbox, checkbox should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Form Attributes 1 Understanding Define TextBox, ComboBox, ListBox, CheckBox means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എന്നിരിക്കട്ടെ Explain Form Attributes 1 Understanding Define TextBox, ComboBox, ListBox, CheckBox എന്നിവയുടെ definition/meaning എന്നിവയുടെ. എന്നിവയുടെ steps, application എന്നിവയുടെ. Technical terms English-എന്നിരിക്കട്ടെ എന്നിവയുടെ.

2 Understanding and RadioButton

2 Understanding and RadioButton is an official topic in Module 3 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding and RadioButton using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding and radiobutton should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding and RadioButton means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding and RadioButton

റേഡിയോബട്ടൺ റേഡിയോ definition/meaning റേഡിയോബട്ടൺ. റേഡിയോ റേഡിയോ റേഡിയോ,
steps, application റേഡിയോ റേഡിയോബട്ടൺ റേഡിയോ. Technical terms English- റേഡിയോ
റേഡിയോബട്ടൺ.

Define Timer and explain Menu Creation

Define Timer and explain Menu Creation is an official topic in Module 3 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define Timer and explain Menu Creation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define timer and explain menu creation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define Timer and explain Menu Creation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-്കു Define Timer and explain Menu Creation

മെനു സൃഷ്ടിക്കുന്നതിനുള്ള ടൈമർ നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. ടൈമർ സൃഷ്ടിക്കുന്നതിനുള്ള ടൈമർ, steps, application ടൈമർ സൃഷ്ടിക്കുന്നതിനുള്ള ടൈമർ. Technical terms English-ൽ ടൈമർ സൃഷ്ടിക്കുന്നതിനുള്ള ടൈമർ.

Explain Event Driven Programming 1 Understanding Understand Form Events - Load, Activate, Close,

Explain Event Driven Programming 1 Understanding Understand Form Events - Load, Activate, Close, is an official topic in Module 3 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Event Driven Programming 1 Understanding Understand Form Events - Load, Activate, Close, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain event driven programming 1 understanding understand form events - load, activate, close, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Event Driven Programming 1 Understanding Understand Form Events - Load, Activate, Close, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എന്നിരിക്കട്ടെ Explain Event Driven Programming 1 Understanding Understand Form Events - Load, Activate, Close, എന്നിവയുടെ definition/meaning എന്നിവയെക്കുറിച്ചുള്ള. എന്നിവയുടെ steps, application എന്നിവയെക്കുറിച്ചുള്ള. Technical terms English-ൽ എന്നിവയെക്കുറിച്ചുള്ള.

Keyboard Events - KeyPress, KeyDown and Mouse 2 Understanding Events - Click, DoubleClick, MouseMove Explain Exception Handling; compare structured

Keyboard Events - KeyPress, KeyDown and Mouse 2 Understanding Events - Click, DoubleClick, MouseMove Explain Exception Handling; compare structured is an official topic in Module 3 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Keyboard Events - KeyPress, KeyDown and Mouse 2 Understanding Events - Click, DoubleClick, MouseMove Explain Exception Handling; compare structured using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, keyboard events - keypress, keydown and mouse 2 understanding events - click, doubleclick, mousemove explain exception handling; compare structured should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Keyboard Events - KeyPress, KeyDown and Mouse 2 Understanding Events - Click, DoubleClick, MouseMove Explain Exception Handling; compare structured means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-☐☐ Keyboard Events - KeyPress, KeyDown and Mouse 2 Understanding Events - Click, DoubleClick, MouseEvent Explain Exception Handling; compare structured ☐☐☐☐☐☐☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐☐☐☐☐☐, steps, application ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms English-☐☐☐☐☐☐☐☐☐☐☐☐☐☐.

1 Understanding and unstructured exception handling.

1 Understanding and unstructured exception handling. is an official topic in Module 3 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding and unstructured exception handling. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding and unstructured exception handling. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding and unstructured exception handling. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 1 Understanding and unstructured exception handling. പദപരിച്ഛേദനം definition/meaning പദപരിച്ഛേദനം. പദപരിച്ഛേദനം പദപരിച്ഛേദനം, steps, application പദപരിച്ഛേദനം പദപരിച്ഛേദനം. Technical terms English-പദപരിച്ഛേദനം പദപരിച്ഛേദനം.

Explain Crystal Reports 1 Understanding Define Database Fields, special fields, summary

Explain Crystal Reports 1 Understanding Define Database Fields, special fields, summary is an official topic in Module 3 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Crystal Reports 1 Understanding Define Database Fields, special fields, summary using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain crystal reports 1 understanding define database fields, special fields, summary should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Crystal Reports 1 Understanding Define Database Fields, special fields, summary means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എന്നു പറയുന്ന Crystal Reports 1 Understanding Define Database Fields, special fields, summary ഫീൽഡ് definition/meaning എഴുതുക. ഫീൽഡ് ഫീൽഡ് ഫീൽഡ്, steps, application ഫീൽഡ് ഫീൽഡ്. Technical terms English-എന്നു പറയുന്ന ഫീൽഡ് ഫീൽഡ്.

2 Understanding fields, lines and boxes Define Report Header and Report Footer, Page

2 Understanding fields, lines and boxes Define Report Header and Report Footer, Page is an official topic in Module 3 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding fields, lines and boxes Define Report Header and Report Footer, Page using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding fields, lines and boxes define report header and report footer, page should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding fields, lines and boxes Define Report Header and Report Footer, Page means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-2 Understanding fields, lines and boxes Define Report Header and Report Footer, Page definition/meaning. steps, application. Technical terms English-

Header and Page Footer, Group Header and Group 3 Understanding Footer

Header and Page Footer, Group Header and Group 3 Understanding Footer is an official topic in Module 3 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Header and Page Footer, Group Header and Group 3 Understanding Footer using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, header and page footer, group header and group 3 understanding footer should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Header and Page Footer, Group Header and Group 3 Understanding Footer means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-Header and Page Footer, Group Header and Group 3 Understanding Footer
 definition/meaning
 . steps, application
 . Technical terms English-

Define Formatting Fields

Define Formatting Fields is an official topic in Module 3 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define Formatting Fields using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define formatting fields should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define Formatting Fields means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Define Formatting Fields
 definition/meaning
 definition/meaning
, steps, application

 Technical terms English-

Define Printer and Page Settings

Define Printer and Page Settings is an official topic in Module 3 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define Printer and Page Settings using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define printer and page settings should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define Printer and Page Settings means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Define Printer and Page Settings
definition/meaning
application
Technical terms English-

Describe how to create crystal reports

Describe how to create crystal reports is an official topic in Module 3 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe how to create crystal reports using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe how to create crystal reports should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe how to create crystal reports means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓരോ ക്രിസ്റ്റൽ റിപ്പോർട്ടിന്റെയും വിവരണം നിങ്ങൾക്ക് നൽകേണ്ടതാണ്. നിങ്ങൾക്ക് റിപ്പോർട്ടിന്റെ, steps, application എന്നിവയെക്കുറിച്ചും വിവരണം നൽകേണ്ടതാണ്. Technical terms English-ൽ നിങ്ങൾക്ക് ഉപയോഗിക്കേണ്ടതാണ്.

List advantages of crystal report 1 Understanding Window Forms - Form Attributes - TextBox - Label - Combo Box - List Box - Check Box - Radio Button - Timer - Menu - Dialog Boxes - MsgBox - Input Box - Events - Form Events - Keyboard Events - Mouse Events - Exceptions - Try Catch - Crystal Reports - Database Fields - Special Fields - Summary Fields - Lines - Boxes - Report Header - Report Footer - Page Header - Page Footer - Group Header - Group Header - Formatting Fields - Printer Settings - Page Settings.

List advantages of crystal report 1 Understanding Window Forms - Form Attributes - TextBox - Label - Combo Box - List Box - Check Box - Radio Button - Timer - Menu - Dialog Boxes - MsgBox - Input Box - Events - Form Events - Keyboard Events - Mouse Events - Exceptions - Try Catch - Crystal Reports - Database Fields - Special Fields - Summary Fields - Lines - Boxes - Report Header - Report Footer - Page Header - Page Footer - Group Header - Group Header - Formatting Fields - Printer Settings - Page Settings. is an official topic in Module 3 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify List advantages of crystal report 1 Understanding Window Forms - Form Attributes - TextBox - Label - Combo Box - List Box - Check Box - Radio Button - Timer - Menu - Dialog Boxes - MsgBox - Input Box - Events - Form Events - Keyboard Events - Mouse Events - Exceptions - Try Catch - Crystal Reports - Database Fields - Special Fields - Summary Fields - Lines - Boxes - Report Header - Report Footer - Page Header - Page Footer - Group Header - Group Header - Formatting Fields - Printer Settings. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list advantages of crystal report 1 understanding window forms - form attributes - textbox - label - combo box - list box - check box - radio button - timer - menu - dialog boxes - msgbox - input box - events - form events - keyboard events - mouse events - exceptions - try catch - crystal reports - database fields - special fields - summary fields - lines - boxes - report header - report footer - page header - page footer - group header - group header - formatting fields - printer settings - page settings. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What List advantages of crystal report 1 Understanding Window Forms - Form Attributes - TextBox - Label - Combo Box - List Box - Check Box - Radio Button - Timer - Menu - Dialog Boxes - MsgBox - Input Box - Events - Form Events - Keyboard Events - Mouse Events - Exceptions - Try Catch - Crystal Reports - Database Fields - Special Fields - Summary Fields - Lines - Boxes - Report Header - Report Footer - Page Header - Page Footer - Group Header - Group Header - Formatting Fields - Printer Settings - Page Settings. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- List advantages of crystal report 1 Understanding Window Forms - Form Attributes - TextBox - Label - Combo Box - List Box - Check Box - Radio Button - Timer - Menu - Dialog Boxes - MsgBox - Input Box - Events - Form Events - Keyboard Events - Mouse Events - Exceptions - Try Catch - Crystal Reports - Database Fields - Special Fields - Summary Fields - Lines - Boxes - Report Header - Report Footer - Page Header - Page Footer - Group Header - Group Header - Formatting Fields - Printer Settings - Page Settings. definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Explain the concept of windows programming, events and exceptions.

M3.01	Describe the procedure of creating Window Forms	1
Understanding		
M3.02	Explain Form Attributes	1
Understanding		
M3.03	Define TextBox, ComboBox, ListBox, CheckBox and RadioButton	2
Understanding		
M3.04	Define Timer and explain Menu Creation	1
Understanding		
M3.05	Explain Event Driven Programming	1
Understanding		
M3.06	Understand Form Events - Load, Activate, Close, Keyboard Events - KeyPress, KeyDown and Mouse Events - Click, DoubleClick, MouseMove	2
Understanding		
M3.07	Explain Exception Handling; compare structured and unstructured exception handling.	1
Understanding		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Outline ADO.NET and ASP.NET.

This module is presented from the official syllabus scope for Visual Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline ADO.NET and ASP.NET.
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

Understand ADO.NET

Understand ADO.NET is an official topic in Module 4 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand ADO.NET using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand ado.net should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand ADO.NET means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Understand ADO.NET ഏതെങ്കിലും ഒരു ഡാറ്റാബേസ് നോട്ടീഫിക്കേഷൻ/definition/meaning നൽകുക. ഏതെങ്കിലും ഡാറ്റാബേസ് നോട്ടീഫിക്കേഷൻ, steps, application നൽകുക. Technical terms English-ൽ നൽകുക.

Define Connection Object and Command Object 2 Understanding Define DataAdapter, DataSet, DataTable and

Define Connection Object and Command Object 2 Understanding Define DataAdapter, DataSet, DataTable and is an official topic in Module 4 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define Connection Object and Command Object 2 Understanding Define DataAdapter, DataSet, DataTable and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define connection object and command object 2 understanding define dataadapter, dataset, datatable and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define Connection Object and Command Object 2 Understanding Define DataAdapter, DataSet, DataTable and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നു Define Connection Object and Command Object 2 Understanding Define DataAdapter, DataSet, DataTable and ഏകദേശം നിർവ്വചനം definition/meaning നിർവ്വചനം. നിർവ്വചനം നിർവ്വചനം നിർവ്വചനം, steps, application നിർവ്വചനം നിർവ്വചനം നിർവ്വചനം. Technical terms English-എന്നു നിർവ്വചനം നിർവ്വചനം.

3 Understanding DataRow

3 Understanding DataRow is an official topic in Module 4 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding DataRow using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding datarow should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding DataRow means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-3 Understanding DataRow ഫോർമാറ്റിംഗ് ഓപ്പറേഷൻ definition/meaning ഫോർമാറ്റിംഗ് ഓപ്പറേഷൻ. ഫോർമാറ്റിംഗ് ഓപ്പറേഷൻ, steps, application ഫോർമാറ്റിംഗ് ഓപ്പറേഷൻ ഓപ്പറേഷൻ. Technical terms English-3 ഫോർമാറ്റിംഗ് ഓപ്പറേഷൻ.

List advantages of ADO.NET

List advantages of ADO.NET is an official topic in Module 4 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify List advantages of ADO.NET using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list advantages of ado.net should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What List advantages of ADO.NET means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- List advantages of ADO.NET
 definition/meaning .
 application
 . Technical terms English-
 .

Explain ASP .NET

Explain ASP .NET is an official topic in Module 4 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain ASP .NET using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain asp .net should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain ASP .NET means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain ASP .NET definition/meaning . steps, application . Technical terms English-

Understand to design Web Pages using ASP .NET

Understand to design Web Pages using ASP .NET is an official topic in Module 4 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand to design Web Pages using ASP .NET using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand to design web pages using asp .net should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand to design Web Pages using ASP .NET means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Understand to design Web Pages using ASP .NET
 definition/meaning .
 , steps, application . Technical terms English-
 .

List ASP .NET Controls

List ASP .NET Controls is an official topic in Module 4 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify List ASP .NET Controls using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list asp .net controls should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What List ASP .NET Controls means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-List ASP .NET Controls നു് നി്വർത്തനം നു് നി്വർത്തനം definition/meaning നു് നി്വർത്തനം. നു് നി്വർത്തനം നു് നി്വർത്തനം, steps, application നു് നി്വർത്തനം നു് നി്വർത്തനം നു് നി്വർത്തനം. Technical terms English-നു് നി്വർത്തനം നു് നി്വർത്തനം.

Explain Web Forms 2 Understanding ADO .NET - Architecture - Connection - Command - Data Adapter - DataSet - DataTable - DataRow - ASP .NET - Forms - Controls. Text / Reference: T/R Book Title/Author R1 Introduction to Visual basic.NET- NIIT, Prentice Hall of India , 2005 R2 David S. Platt, Introducing Microsoft .NET, Microsoft Press Publishers R3 Introduction to Microsoft ASP.NET Work Book, Microsoft Press R4 Danny Ryan & Tommy Ryan, ASP.NET, Wiley Online Resources: Sl.No Website Link 1 <https://www.tutorialspoint.com/vb.net> 2 <https://www.thecodingguys.net/tutorials/visualbasic/vb-tutorial> 3 www.visual-basic-tutorials.com 4 <https://www.vbtutor.net/vbtutor.html> 5 <https://www.vbtutor.net/index.php/visual-basic-2017-tutorial>

Explain Web Forms 2 Understanding ADO .NET - Architecture - Connection - Command - Data Adapter - DataSet - DataTable - DataRow - ASP .NET - Forms - Controls. Text / Reference: T/R Book Title/Author R1 Introduction to Visual basic.NET- NIIT, Prentice Hall of India , 2005 R2 David S. Platt, Introducing Microsoft .NET, Microsoft Press Publishers R3 Introduction to Microsoft ASP.NET Work Book, Microsoft Press R4 Danny Ryan & Tommy Ryan, ASP.NET, Wiley Online Resources: Sl.No Website Link 1 <https://www.tutorialspoint.com/vb.net> 2 <https://www.thecodingguys.net/tutorials/visualbasic/vb-tutorial> 3 www.visual-basic-tutorials.com 4 <https://www.vbtutor.net/vbtutor.html> 5 <https://www.vbtutor.net/index.php/visual-basic-2017-tutorial> is an official topic in Module 4 of Visual Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Web Forms 2 Understanding ADO .NET - Architecture - Connection - Command - Data Adapter - DataSet - DataTable - DataRow - ASP .NET - Forms - Controls. Text / Reference: T/R Book Title/Author R1 Introduction to Visual basic.NET- NIIT, Prentice Hall of India , 2005 R2 David S. Platt, Introducing Microsoft .NET, Microsoft Press Publishers R3 Introduction to Microsoft ASP.NET Work Book, Microsoft Press R4 Danny Ryan & Tommy Ryan, ASP.NET, Wiley Online Resources: Sl.No Website Link 1 <https://www.tutorialspoint.com/vb.net> 2 <https://www.thecodingguys.net/tutorials/visualbasic/vb-tutorial> 3 [www.visual-basic-](http://www.visual-basic-tutorials.com)

tutorials.com 4 <https://www.vbtutor.net/vbtutor.html> 5

<https://www.vbtutor.net/index.php/visual-basic-2017-tutorial> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain web forms 2 understanding ado .net - architecture - connection - command - data adapter - dataset - datatable - datarow - asp .net - forms - controls. text / reference: t/r book title/author r1 introduction to visual basic.net- niit, prentice hall of india , 2005 r2 david s. platt, introducing microsoft .net, microsoft press publishers r3 introduction to microsoft asp.net work book, microsoft press r4 danny ryan & tommy ryan, asp.net, wiley online resources: sl.no website link 1

<https://www.tutorialspoint.com/vb.net> 2

<https://www.thecodingguys.net/tutorials/visualbasic/vb-tutorial> 3 www.visual-basic-tutorials.com 4 <https://www.vbtutor.net/vbtutor.html> 5

<https://www.vbtutor.net/index.php/visual-basic-2017-tutorial> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Web Forms 2 Understanding ADO .NET - Architecture - Connection - Command - Data Adapter - DataSet - DataTable - DataRow - ASP .NET - Forms - Controls. Text / Reference: T/R Book Title/Author R1 Introduction to Visual basic.NET- NIIT, Prentice Hall of India , 2005 R2 David S. Platt, Introducing Microsoft .NET, Microsoft Press Publishers R3 Introduction to Microsoft ASP.NET Work Book, Microsoft Press R4 Danny Ryan & Tommy Ryan, ASP.NET, Wiley Online Resources: Sl.No Website Link 1 https://www.tutorialspoint.com/vb.net 2 https://www.thecodingguys.net/tutorials/visualbasic/vb-tutorial 3 www.visual-basic-tutorials.com 4 https://www.vbtutor.net/vbtutor.html 5 https://www.vbtutor.net/index.php/visual-basic-2017-tutorial means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain Web Forms 2 Understanding ADO .NET - Architecture - Connection - Command - Data Adapter - DataSet - DataTable - DataRow - ASP .NET - Forms - Controls. Text / Reference: T/R Book Title/Author R1 Introduction to Visual basic.NET- NIIT, Prentice Hall of India , 2005 R2 David S. Platt, Introducing Microsoft .NET, Microsoft Press Publishers R3 Introduction to Microsoft ASP.NET Work Book, Microsoft Press R4 Danny Ryan & Tommy Ryan, ASP.NET, Wiley Online Resources: Sl.No Website Link 1 <https://www.tutorialspoint.com/vb.net> 2 <https://www.thecodingguys.net/tutorials/visualbasic/vb-tutorial> 3 www.visual-basic-tutorials.com 4 <https://www.vbtutor.net/vbtutor.html> 5 <https://www.vbtutor.net/index.php/visual-basic-2017-tutorial> Visual Basic .NET definition/meaning. Visual Basic .NET, steps, application. Technical terms English-Visual Basic .NET.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Outline ADO.NET and ASP.NET.

M4.01	Understand ADO.NET Understanding	1
M4.02	Define Connection Object and Command Object Understanding	2
M4.03	Define DataAdapter, DataSet, DataTable and Understanding DataRow	3
M4.04	List advantages of ADO.NET Understanding	1
M4.05	Explain ASP .NET Understanding	1
M4.06	Understand to design Web Pages using ASP .NET Understanding	3
M4.07	List ASP .NET Controls Understanding	2
M4.08	Explain Web Forms Understanding	2

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Define Visual Programming 1 Understanding List advantages and disadvantages of visual. (3 marks)

Answer: Begin with the standard definition or identity of *Define Visual Programming 1 Understanding List advantages and disadvantages of visual*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 1 Understanding programming Familiarize .NET Framework and explain .NET. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding programming Familiarize .NET Framework and explain .NET*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Understanding Framework architecture. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding Framework architecture*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Understand CLR and .Net Class Library. (3 marks)

Answer: Begin with the standard definition or identity of *Understand CLR and .Net Class Library*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Understand If - Then, If - Then - Else Statements. (3 marks)

Answer: Begin with the standard definition or identity of *Understand If - Then, If - Then - Else Statements*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Understand Select Case Statement. (3 marks)

Answer: Begin with the standard definition or identity of *Understand Select Case Statement*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Understand Do While - Loop Statement. (3 marks)

Answer: Begin with the standard definition or identity of *Understand Do While - Loop Statement*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Understand For - Next Statement. (3 marks)

Answer: Begin with the standard definition or identity of *Understand For - Next Statement*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain Describe the procedure of creating Window Forms. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the procedure of creating Window Forms*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain Form Attributes 1 Understanding Define TextBox, ComboBox, ListBox, CheckBox. (3 marks)

Answer: Begin with the standard definition or identity of *Explain Form Attributes 1 Understanding Define TextBox, ComboBox, ListBox, CheckBox*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Understanding and RadioButton. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding and RadioButton*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Define Timer and explain Menu Creation. (3 marks)

Answer: Begin with the standard definition or identity of *Define Timer and explain Menu Creation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain Understand ADO.NET. (3 marks)

Answer: Begin with the standard definition or identity of *Understand ADO.NET*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Define Connection Object and Command Object 2 Understanding Define DataAdapter, DataSet, DataTable and. (3 marks)

Answer: Begin with the standard definition or identity of *Define Connection Object and Command Object 2 Understanding Define DataAdapter, DataSet, DataTable and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding DataRow. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding DataRow*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain List advantages of ADO.NET. (3 marks)

Answer: Begin with the standard definition or identity of *List advantages of ADO.NET*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Define Visual Programming 1 Understanding List advantages and disadvantages of visual.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Understand If - Then, If - Then - Else Statements**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Describe the procedure of creating Window Forms**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Understand ADO.NET**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.tutorialspoint.com/vb.net>
- <https://www.thecodingguys.net/tutorials/visualbasic/vb-tutorial> www.visual-basic-tutorials.com <https://www.vbtutor.net/vbtutor.html>

- <https://www.vbtutor.net/index.php/visual-basic-2017-tutorial>

IN-PAGE SEARCH

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